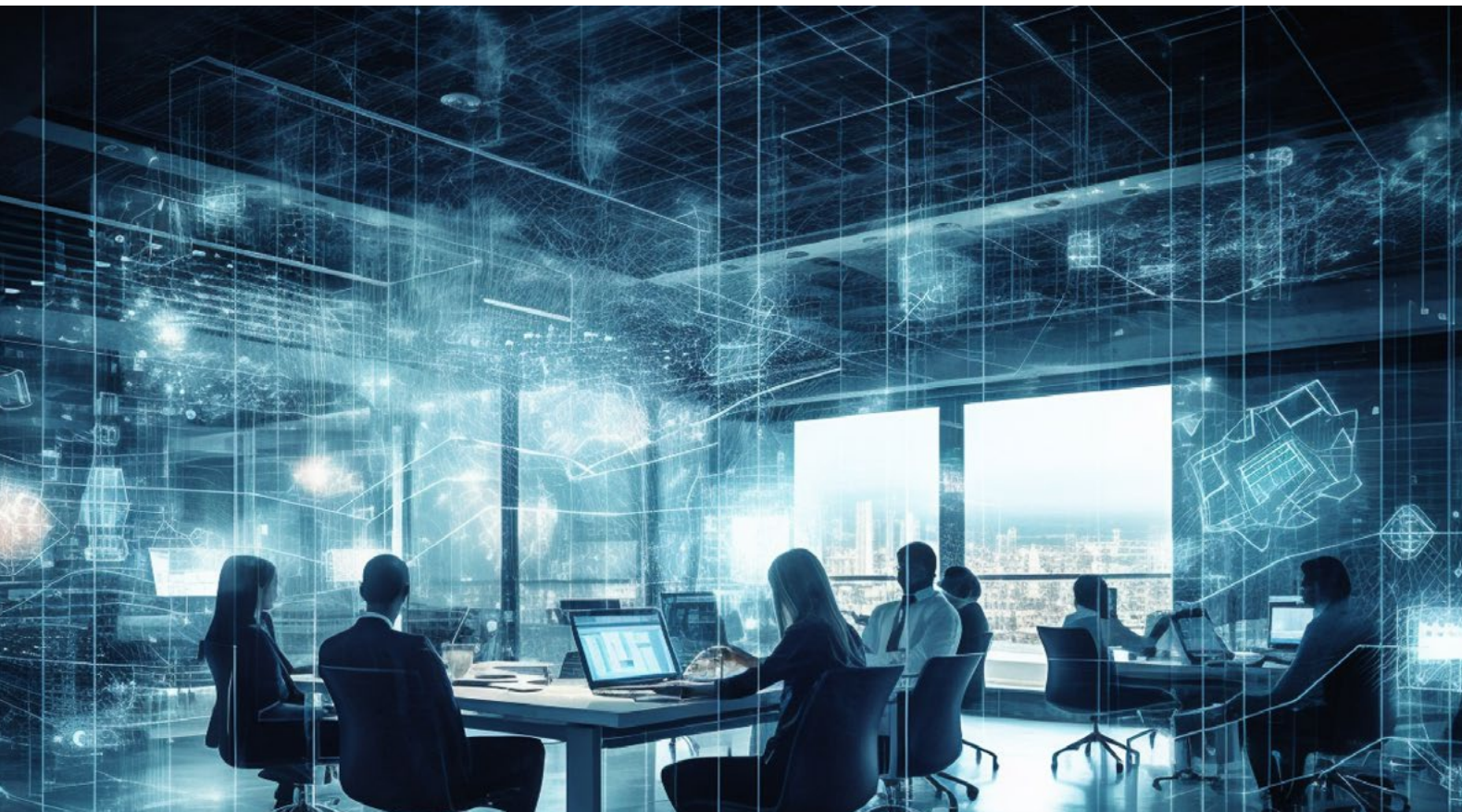


An introduction to

THE DATA COOPERATION CANVAS



A CANVAS MODEL TO DESCRIBE EXISTING AND
EXPLORE NEW DATA COOPERATIONS

Ron van der Lans
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www.datacooperationcanvas.eu



DATA SPACE FOR
SMART AND SUSTAINABLE
CITIES AND COMMUNITIES



braxwell



**DATA SPACE FOR
SMART AND SUSTAINABLE
CITIES AND COMMUNITIES**



**City of
Amsterdam**

The Data Cooperation Canvas has been conceived as part of the preparatory actions for the Data Space for Smart and Sustainable Cities and Communities (DS4SSCC), part of the Digital Europe Program of the European Commission. The canvas was developed by Ron van der Lans and Jasper Soetendal of Braxwell.com in the role of external experts strategic data partnerships of the Directorate Digitalization & Innovation of the City of Amsterdam and has been added by other participants of DS4SSCC working groups.





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WHAT IS THE DATA COOPERATION CANVAS?

The Data Cooperation Canvas is much like the well-known 'Business Model Canvas'.

Like the Business Model Canvas is used to describe and explore business models, the Data Cooperation Canvas can be used to describe and explore data cooperations.



GUIDANCE FOR A SUCCESSFUL DATA COOPERATION

Over the last years we have been involved in dozens of (potential) data cooperations, ranging from highly successful to miserably failed.

Through this trial and error we have gained valuable knowledge and experience on how to set up a successful data cooperation. We want to share these insights, structured in a clear and helpful tool.

This is how The Data Cooperation Canvas was born, as part of the preparatory actions for the Data Space for Smart and Sustainable Cities and Communities (DS4SSCC).

With this book we hope to pass on this knowledge and experience within and beyond DS4SSCC, providing guidance for successful data cooperations.

**Ron van der Lans &
Jasper Soetendal**
Authors

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Why do you need the canvas?

A shared picture of the opportunity, a common understanding on possible solutions and a joint vision on how to move forward are vital aspects of successful cooperation. Clear communication serves as a fundamental prerequisite for achieving these goals. As the number of discussions and written materials grows, the necessity for a shared and well-defined framework becomes increasingly apparent.

Enter the Data Cooperation Canvas: a tool that compels conciseness, clarity, and completeness. As it is a one-pager, representing merely the tip of the iceberg in the vast sea of paperwork essential for a successful data cooperation, it functions as:

- A top-level shared understanding for all key partners, to structure discussions internally
- A tool to explore sustainable business models, revenue streams and governance mechanisms for the data cooperation
- A valuable framework for effectively communicating the scope of the cooperation to external stakeholders



Explore new data
cooperations



Describe existing
data cooperations



Share a common
picture



Cover all important
aspects

THE CANVAS AND ITS ELEMENTS

A new or existing data cooperation can be described by 14 essential building blocks in 3 areas.

In this document we will zoom in on each element and provide examples and models to fill these elements in your data cooperation canvas.

Why?



Context



Added value



Motivation & objectives

Organizational



Key partners



Resources



Shared processes



Business case



Governance model



Implementation roadmap



Current status

Technical



Data & data sources



Interoperability



Technical concepts/models



Infrastructure characteristics

Organizational

Key partners

Who are the partners involved in the data exchange? What are their roles?



Resources

What organizational resources are required for this data cooperation? What resources are available already? What needs to be done to get all required resources?



Business case

What are the costs of the data exchange? Who is paying? What are the revenues? Who is profiting? What compensation, fees or other financials are needed?



Governance model

How are rules, norms and actions structured/sustained/regulated to control the data exchange?



Shared processes

What steps are performed as a shared process in the data exchange? What steps are done individually?

	Individual	Shared
Use	☐	☐
↑		
Visualize	☐	☐
↑		
Interpret	☐	☐
↑		
Combine	☐	☐
↑		
Transform	☐	☐
↑		
Store	☐	☐
↑		
Create	☐	☐



Why?

Context

What is the business context that creates the opportunity/necessity for data exchange?



Added value

Why will this data cooperation succeed? What is the added value for participants?



Motivation & objectives

What is the motivation for the key partners to join the data exchange? What are their main objectives of participating?



Technical

Data & data sources

What data is exchanged? What data sources are used?



Interoperability

How can the data be uniformed/standardized/combined? What shared concepts, languages, formats, or methods can be used? Is it hard to combine all the data? Or are standard definitions available? What data standards & formats are used or need to be used?



Implementation roadmap

What approach will be used for realizing and implementing the data exchange?



Technical concepts/models

What technical concepts or models need to be in place for the data exchange. What MIMs are implemented and how are they implemented?



Infrastructure characteristics

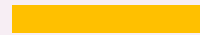
What technical infrastructure is needed for the data exchange?

- What cloud/server infrastructure is used
- What technology stack is used
- What standard software is used?
- In-house development or external parties?
- Central/decentral/distributed model
- How can be connected (API, feeds, downloads, etc.)



Current status

What is the current status of the cooperation?



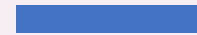
Exploratory stage



Preparatory stage



Implementation stage



Operational stage



Scaling stage



HOW TO USE THE DATA COOPERATION CANVAS

For exploration, the canvas can be used as a blue print on how a **potential** data cooperation can be 'build'.

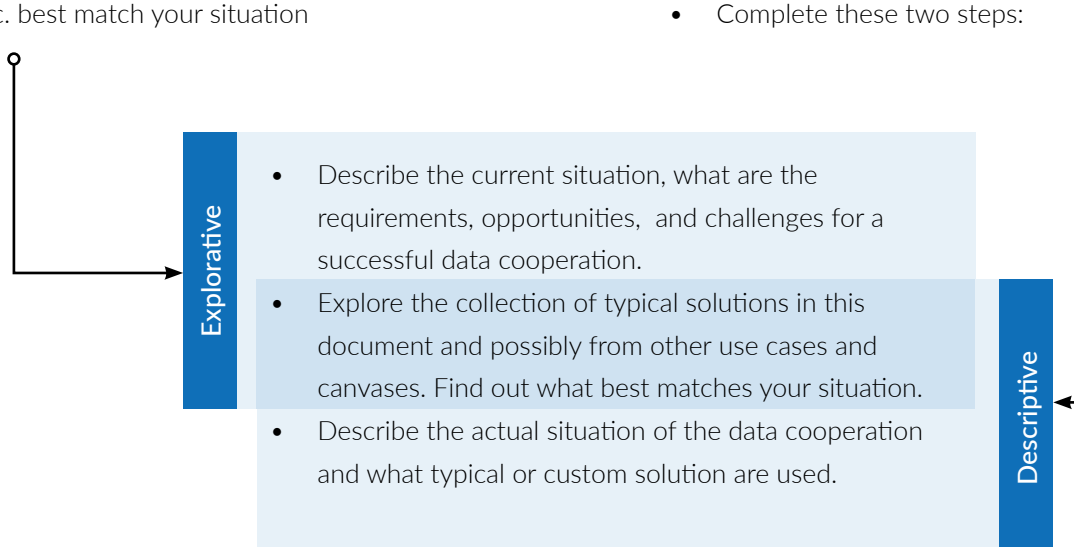
For **existing** data cooperations, the canvas can be used as a template to describe the current situation.

Explorative

- Before or while setting up a data cooperation
- To describe the current situation, requirements, opportunities and challenges
- To explore typical solutions, being inspired by other descriptive data cooperation canvases and figuring out what governance structures, business model/revenue streams, etc. best match your situation
- Complete these two steps:

Descriptive

- After setting up a data cooperation
- As a structure to describe the actual situation of the data cooperation
- To define the typical solutions that are being used and/or custom solutions that were needed.
- Complete these two steps:



GET STARTED!

To get started with the Data Cooperation Canvas:



1

PRINT AN EMPTY CANVAS

Print the empty template of the Data Cooperation Canvas on page 61 of this book. Fill in the name of the Data Cooperation on the top.



2

FILL ALL ELEMENTS

Fill all elements of the canvas, using the suggested actions that you'll find for each element.



3

COLLECT ADDITIONAL INFORMATION

Using the suggested additional actions for all elements, building a document with more detailed information

And.... action!

For each element of the canvas, we suggest actions to fill in that area. These are marked with these icons:



Action

An instruction of actions that should be taken to complete the canvas



Additional

Additional actions that can be taken. These will not fit on the one-page canvas, but will be helpful nevertheless.

Why?

In the red 'Why?' part we specify the context that creates the opportunity or necessity for a data cooperation, how value can be added and what the motivation and objectives of participants are.



Context

What is the business/organizational context that creates the opportunity/necessity for data exchange?



Action:

In a few sentences, describe the current situation, and focus on the problem of the parties involved. What could or should be improved?



Added value

Why will this data cooperation succeed? What is the added value for participants?



Action:

In a few sentences, describe what value is added in the way the (proposed) data cooperation solves or could solve this problem. What parties will profit from this?



Motivation & objectives

What is the motivation for the key partners to join the data exchange?
What are their main objectives of participating?

In a successful data cooperations, all organizations involved profit.
What is the win/win situation in this case? What is driving the participation of all involved parties? Do all participants have the same motivation and objectives, or do they differ?



Action:

In one sentence, describe the shared motivation to join the data cooperations. Add additional explanation if motivations differ or conflict for key partners.



Additional:

List all (potential) key partners and describe or define their main motivation and objectives for joining the data cooperation. If they are complementary, describe the win/win situation. Reconsider these three 'Why?'-elements if they are conflicting or mutually exclusive.



Key partners



Who are the partners involved in the data exchange? What are their roles?



To create a cooperation, cooperating partners are needed. And while a cooperation can be initiated by one party, a successful data cooperation needs multiple partners working together.

By 'key partners' we mean those organizations, companies or people that are already actively involved in the data cooperation, or are essential for a successful data cooperation in the longer term. In the list on the right you'll find a non-exhaustive list of typical roles for key partners.

Action:

List all key partners that are actively involved in the data cooperation or should be involved. Define what role they (should) have.

Additional:

Besides the 'key partners' that are actively involved in the data cooperation, there might be other entities that are somehow affected or play a more distant role. Having a list of these players, including their role and impact, can be helpful.

Typical roles of key partners

Typical roles of key partners are:

Initiator

Coordinator

Participant

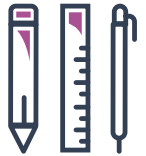
Contributor

User

Financier

Developer





Resources

What organizational resources are required for this data cooperation? What resources are available already? What needs to be done to get all required resources?



The list of typical resources helps to identify resources required for the cooperation to be successful. It might be useful to consider the following questions:

- What expertise/skills/assets required to meet the cooperation objectives can each partner bring?
- What are the gaps in the resources and capabilities required? Could you leverage partners' wider networks?

Action:

List the resources that are needed to setup and run the data cooperation. Start with the most important and add the could-have's or nice-to-have's at the bottom of the list.

Additional:

For each resource, define whether this is already available. If the resource is available, describe what key partner(s) provide this resource. If a resource is not yet available, define what partners may provide this resource and try to estimate whether it will be easy or hard to get this resource.

Typical resources

A non-exhaustive list of type of resources that may be needed to setup and run a data cooperation:

Cash flow/direct funding

Staff time

Data sets

Infrastructure

Products & services

Access to expertise

Access to networks



Business case



What are the costs of the data exchange? Who is paying? What are the revenues? Who is profiting? What compensation, fees or other financials are needed?



Books can be written about business models and business cases.

For the data cooperation canvas, we limit this to a high level summary of the business case, using the typical business models and business model components from the next pages.

The business case is focused on the long run: it should describe the financial situation for the full-grown data cooperation. If funding is needed to start the data cooperation, mention this as well in the business case.

Action:

In one or two sentences, describe the essential parts of your business case. Use the typical business models and its typical elements from the next pages for inspiration. Describe the most important costs and revenues.

Additional:

Extend the list of key partners and their main motivations and objectives (see: Motivation & objectives), and define for each key partner what's their high level business case: what's in it for them?

Typical business models for data cooperations

Based on the DSSC Starter Kit for Data Space Designers: <https://dssc.eu/space/SK/29523973>

Sharing costs	• Participants share their data to meet a shared requirements (e.g., service, process efficiency, transparency) • Every member saves money and time by sharing the burden
Shared compliance	• Participants share their data to meet shared compliance • Every member saves money and time by sharing the burden
Sharing profit	• Participants share their data to create a shared opportunity • Every members shares in the profit created from sharing the data
Shared access	• Participants team up to provide quality-assured, easy access to data of a domain of common interest (open data, business partner data etc.) • Transaction costs go down for all ecosystem members
Joint innovation	• Innovation can only be realized by participants working together • No single ecosystem member has all the necessary means/data to do it on their own.
Combining forces	• Participants agree that joining forces creates a opportunity to team up against existing/emerging competitors/threads • No single ecosystem member has the necessary resources and commitment to do this alone
Greater common good	• Public and private sector share data for a greater common, societal goal (e.g., climate protection)



Typical elements of a business models for data cooperations

What are the incentives for the data cooperation?

- Earn money
- Create value from combined data
- Development of new data driven products/services
- Resource sharing
- Costs sharing
- Compliance
- Knowledge exchange
- Enabling innovation/joint innovation
- Empowering citizens & communities
- Greater common good

What types of business models are available?

Monetarisisation

- Transaction model
- Subscription based model
- Freemium business model

Collaboration

- Data pooling
- Resource pooling
- Delegation data control

Cooperation

- Data altruism
- Citizen science
- Reciprocity: data for data

What does it look like in practice?

- Data purchase
- Software as a service
- Data as a service: discovery, brokerage, aggregation, etc.
- Marketplace
- Trusted third party
- Data donation
- Open data
- Data commons/cooperatives
- Data trusts
- Personal data services

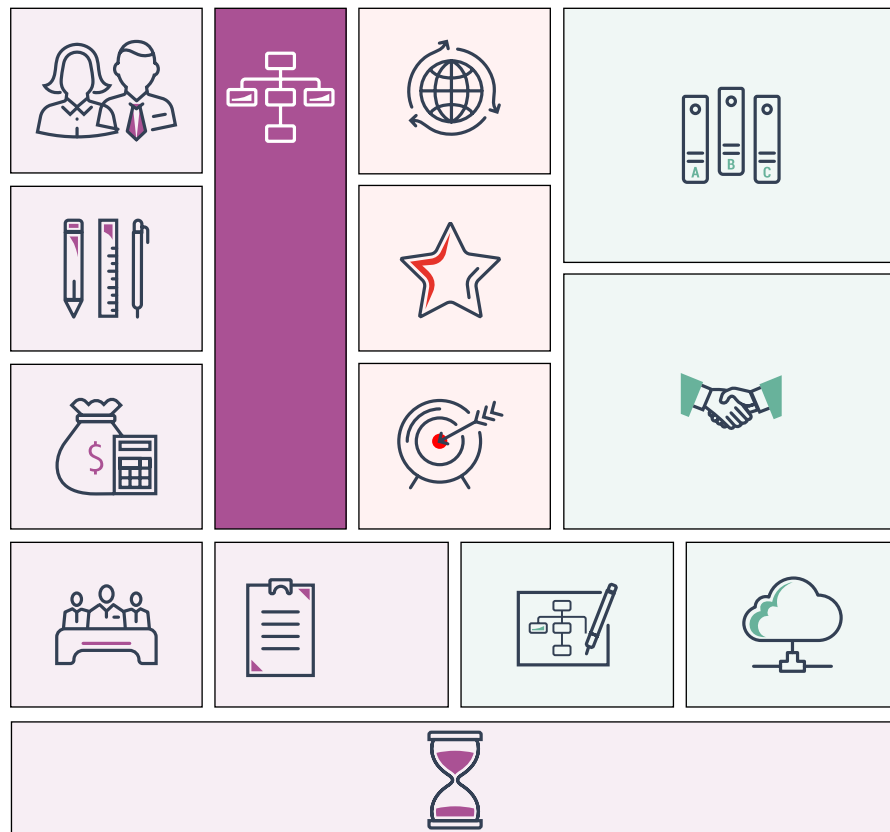
How can it be sustained over time?

- Commercial services/monetisation
- EU public funding
- National/regional public funding
- Membership fee
- Joint investment
- Private investment
- Crowdsourcing
- Donations

Shared processes



What steps are performed as a shared process in the data exchange? What steps are done individually?



From data creator to data user, data flows through different process steps. Although every process is unique, a 'data flow process' can be defined that is common to most cases, see the flow at the right.

In a data cooperation, some of these steps are performed by participants individually, and others are shared/joint in the data cooperation. The defining elements of a data cooperation are the steps performed jointly, which can be identified by checking the boxes in the rightmost columns.

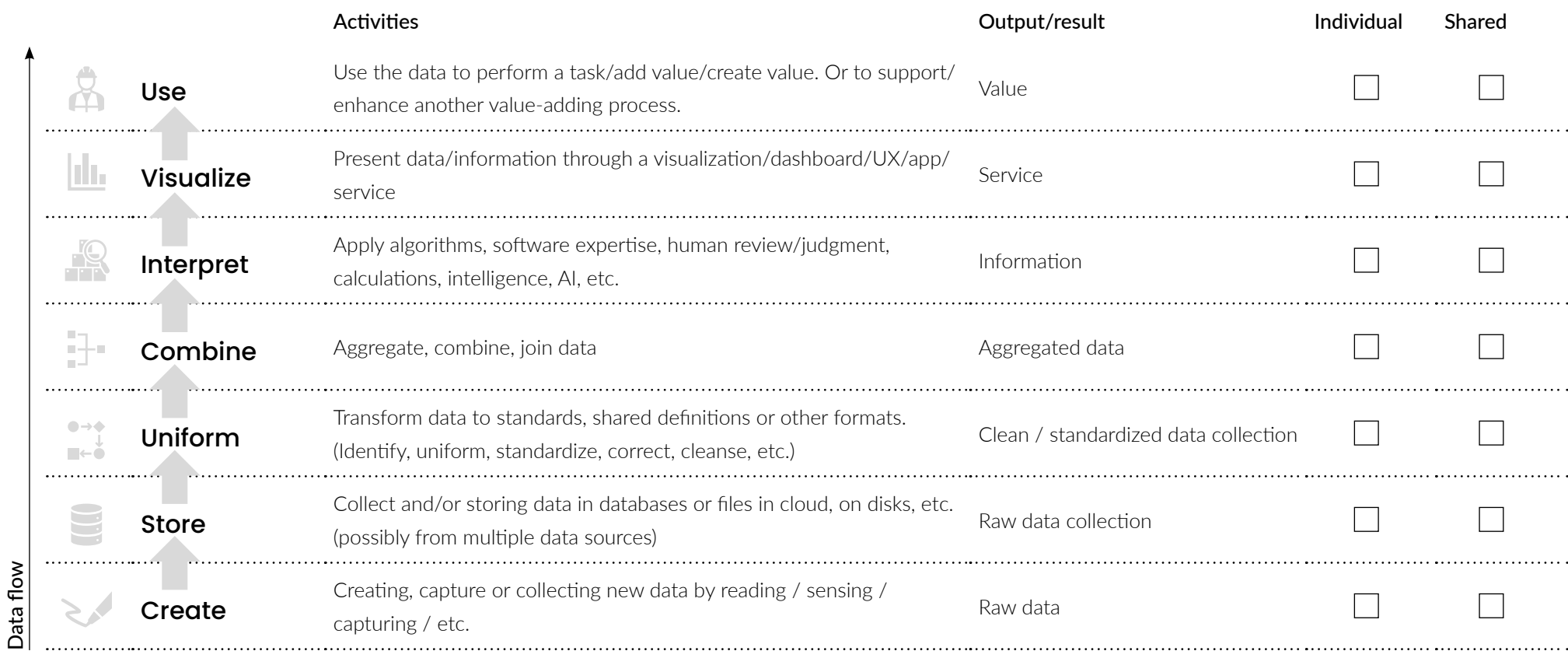
Action:

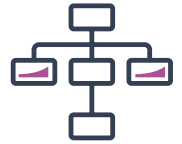
For the 7 steps in the data flow process, tick the box whether each organization is doing the step individually, or it is a joint activity in the data cooperation. Examples of these shared process can be found in the next pages.

Additional:

Think about the implications of having individual or shared process steps. For each step, imagine how the data cooperation would function if the step is a) done individually or b) a joint activity. This may lead to valuable new insights on the data cooperation

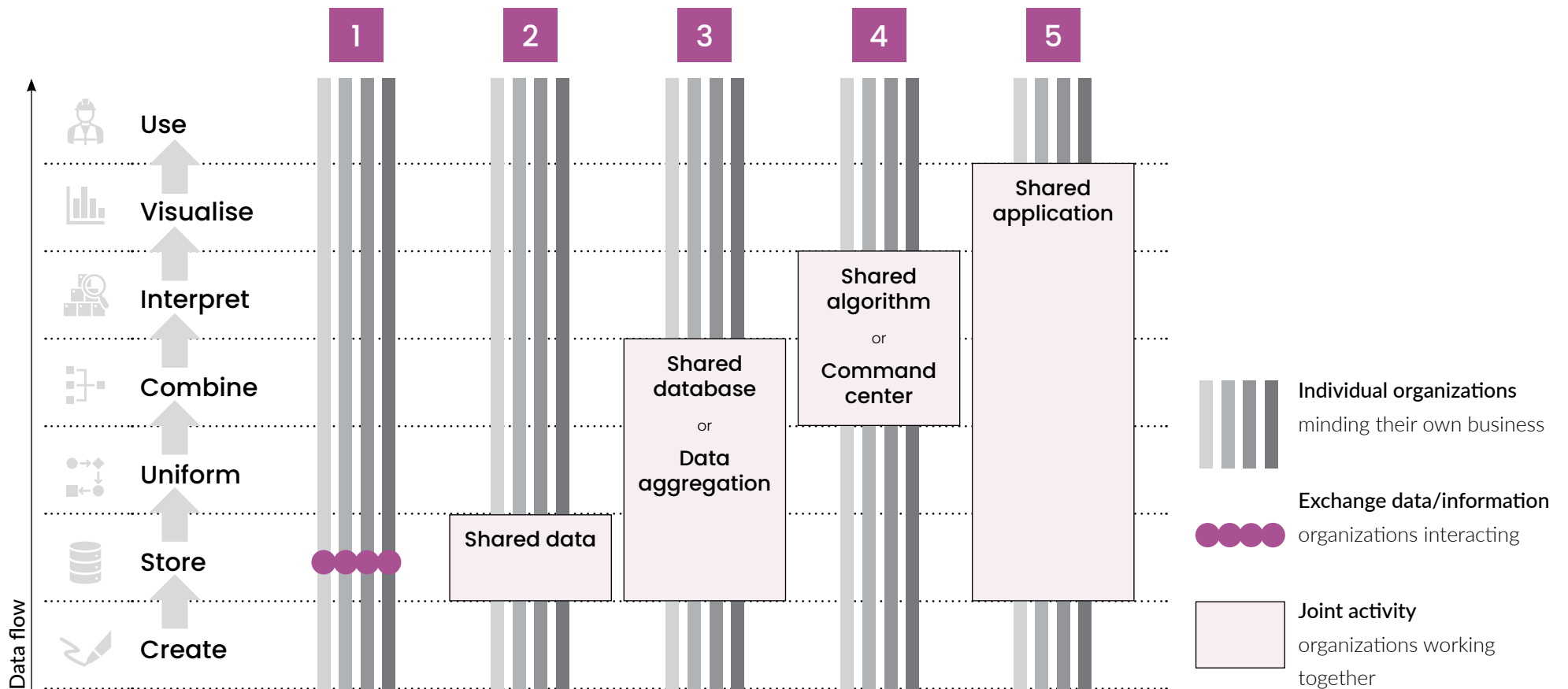
Dataflow proces





Typical shared processes

Below five different examples are shown on how the various parts of the data flow process can be performed individually or can be part of the cooperation. On the next page we will provide an explanation for each example.



	1	2	3	4	5
	Exchanging data	Shared data	Shared database	Shared algorithm	Shared Application
What is shared	Data	Data	Aggregated data	Information	Application
Why?	To exchange data on a 'need to know'-basis, without a shared infrastructure, only data exchanges (like API's)	Share data from others, that is important for you. To use this data within your own domain and systems.	Collect and aggregate data from multiple sources. To use this aggregated data in your own domain and systems.	To share information/interpretation, without the need to share the source data.	Collect and aggregate data from multiple sources, and create a shared application, to fulfill a shared goal.
Example:	Company A requests	Company A, B and C	Company A, B and C	Company A, B and C don't	Company A, B and C
Partner A, B and C are 3 competitors owning multiple parking garages in the city.	Company B: "Do you have available parking spots right now?" using their API. Company B responds: "Yes".	share their visitor numbers data (each in their own format) on a shared drive and use this data in their own systems to create forecasts.	share their visitor numbers data, which are combined and uniformed in one database. Each company uses this aggregated data to create their own forecasts.	want to share their visitor numbers data, because they are competitors. However, they have a shared goal in creating reliable forecasts. They set up an 'shared algorithm', using a trusted third party that combines all their source data, applies an agreed-upon algorithm and outputs the resulting forecasts to all.	create a shared app for visitors. With this app, visitors can book a parking spot, based on actual or predicted availability in all their parkings. They apply an agreed-upon algorithm to direct/suggest specific parking garages to users.

Governance model



How are rules, norms and actions structured/sustained/regulated to control the data exchange?



A governance model specifies the way the data flow process is controlled. It is a set of agreements, policies, structures and operational procedures to specify who can take what decisions in the data cooperation.

On the next page you'll find the typical parts and power structures of a governance model. The following pages provide a set of 11 typical governance models.

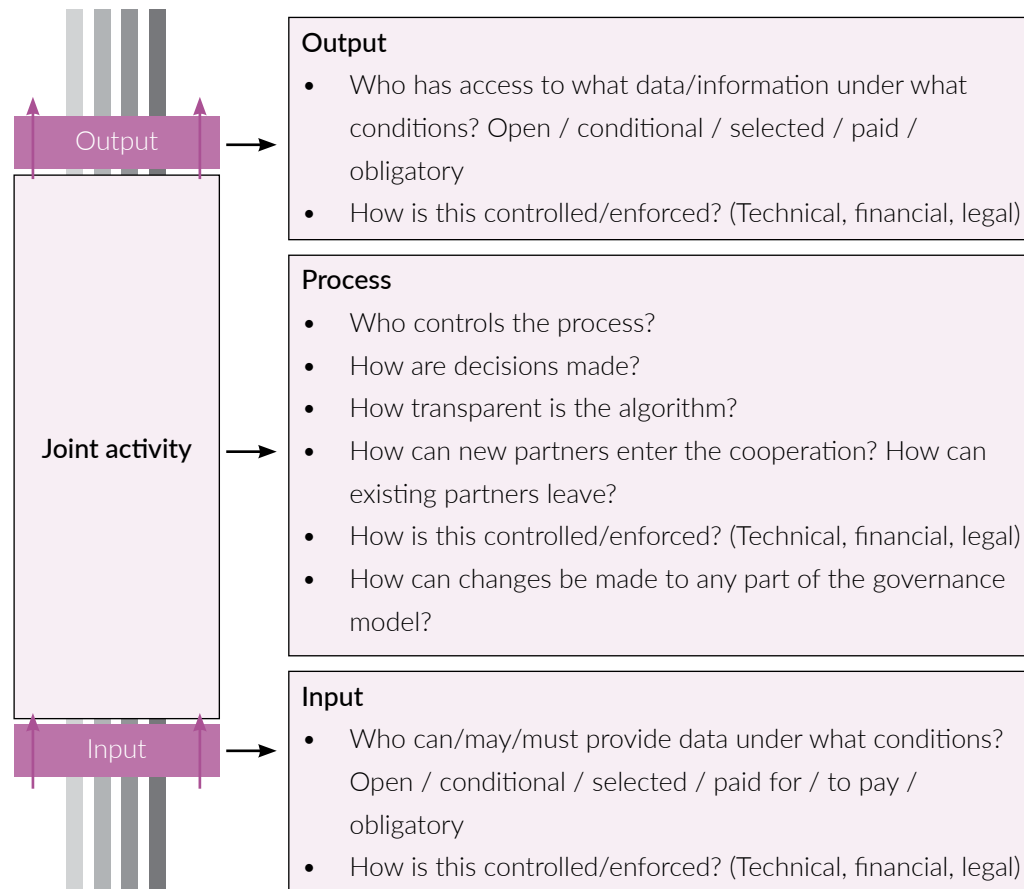
Action:

In a few sentences, describe the essential parts of your governance model. Use the 'typical parts of a governance model' and the 'typical governance models' as guidance.

Additional:

Answer all the questions that are mentioned in the 'Typical parts of a governance model' on the next page.

Typical parts of a governance model

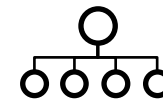


Typical power structures



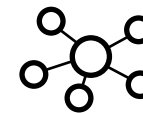
Single

One single entity controls the 'cooperation'.
(Which isn't a cooperation then, is it?)



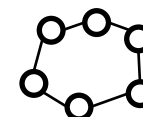
Hierarchical

One entity is in control and coordinates most parts of the cooperation.



Coordinated

One entity initiates and manages the corporation, but all decisions are made by consensus.



Joint

All entities are equal. All decisions are made by consensus.

Typical governance models



Main objective: Enabling re-use and innovation

	Open data / Transparency	Government As A Platform	Data Marketplace	Data Repository
When to use	When one or more entities (government, non-profit or business) want to provide their data to enable re-use and innovation.	When a government or organization wants its (digital) services to be open to anyone, so that users/civil servants, businesses and others can deliver radically better services to the ecosystem/public, more safely, efficiently and accountably.	When data is available, but is spread all over the internet and hard to find. Demand and supply of data are inefficiently matched.	When data is available, but is spread all over the internet and hard to find. Demand and supply of data are inefficiently matched.
How it works	The data is published on a website or portal, is well-documented with metadata and has a license (mostly creative commons or public domain) that allows for a broad use of the data.	The work of an organization/ government is reorganized around a network of shared APIs and components, open-standards and canonical datasets.	A market place provides a platform for data providers to offer their data to potential users. It enables the monetization or brokerage of data for both discovery and transactions between buyers and providers.	A repository provides a listing of available data, offering data providers a way to publish their data and offering data users an efficient way to search for the data they need. The repository provides meta data and a link to the actual data.
Control on input	●●●● High. The initiator decides for itself what data will be published.	●●●● High. The initiator defines the services and components itself.	●●●● Low: if marketplace is open High: if marketplace is curated	●●●● Low: if repository is open High: if repository is curated
Control on use	●○○○ Low. Re-use and innovation for unforeseen applications are welcomed	●○○○ Low. Re-use and innovation for unforeseen applications are welcomed	●○○○ Low. Re-use and innovation for unforeseen applications are welcomed	●○○○ Low. Re-use and innovation for unforeseen applications are welcomed
Examples	- data.europe.eu - data.overheid.nl	- e-Estonia - Jeff Bezos' API Mandate (Amazon)	- Microsoft Azure Data Share - AWS Data Exchange	- Kaggle - Various science data repositories

Partly based on: The Data Economy Lab, <https://tool.thedataeconomylab.com/our-data-models>

Main objective: Cooperate and share costs or profits

	Shared Data	Conditional Access	Data Trust
When to use	When participants have a shared interest or objective, and don't mind to share their (source) data.	When participants don't want to share their (source) data, but can think of very specific situations (like incidents/disasters or bankruptcy) where they have a shared interest or objective to do share the data.	When participants have a shared interest or objective, but won't or can't share their (source) data.
How it works	In a shared data cooperation, participants define what data they can offer, and what data they would like from other participants. If a satisfying set of data can be agreed upon, this data is shared on a shared disk, cloud platform or any other way to exchange this data.	Participants define in detail in what situation what data can be shared to what party. They prepare a technical infrastructure to share the data and agree upon a trusted party to 'ascertain' a situation (like a police/fire department or solicitor) and start the actual process of data sharing.	A trusted third party is used to collect, interpret and combine the data. The third party applies an agreed-upon algorithm or process to provide the required output information. This output information is the minimum amount of (aggregated) data that a) can be shared and b) is needed to fulfill the shared objective.
Control on input	 Medium. All participants agree on what data is input, so there is a strict control. However, only data that participants have and are allowed and willing to share, will be shared. In most cases in the AS-IS format.	 High. It is defined strictly and detailed what data will be shared in what situation	 High. All participants agree in advance what data will be provided.
Control on use	 Medium. Participants agree on what data can be used for what use. However, once the data is out, it is difficult to control the use.	 High. Provided information is tailored to and can only be used for specific objective, as agreed upon in the data cooperation.	 High. Provided information is tailored to and can only be used for specific objective, as agreed upon in the data trust.
Examples	Various B2B initiatives	Various B2B initiatives	Thisisplace.org + various other examples



Typical governance models

Main objective: Protecting valuable data

	Personal control	Data Common
When to use	When persons should have maximum control over the access to their data.	When members of a community want to protect valuable data, so that it can only be used in a way that contributes to the community.
How it works	Providing personal data vaults to people, to store their personal data and to empower maximum user control over data access, permissions and data decision making. To facilitate secure, personal control over your data.	A data common is a digital resource accessible to all members of a community. The data are held in common, through a partnership, a not-for-profit, or another entity, but not owned privately for commercial gain. Members of the common store their own data and define strict conditions for what data can be accessed by what entity [...] Elinor Ostrom. Governing the commons: The evolution of institutions for collective action. Cambridge university press, 1990.
Control on input	●●●● High. Each person has maximum control over its personal data vault	●●●● High. The data common has maximum control over the digital resources.
Control on use	●●●● High. Each person has maximum control over its personal data vault	●●●● High. The data common has maximum control over the digital resources.
Examples	- Digi.me / worlddataexchange.com - Solid (https://solid.mit.edu/)	- UK Biobank - European Open Science Cloud - datacommons.org

	Main objective: Earn money/commercialize data	
	Commercial Data	As-A-Service model
When to use	When data is needed that is not available for free and/or requires a lot of effort to collect, uniform, combine and aggregate.	When a data owner has an interest in adding an extra service layer on top of the available data.
How it works	One or more commercial data providers collect, transform, combine and/or aggregate data and provide the information to users in exchange for a fee (per month, per data set, per GB, etc.).	Instead of providing data to a user, a tool, software, or a service is provided to a user. This tool fulfills the need of the user, who is willing to pay a fee for it.
Control on input	●●●● High. The data provider decides on what data is used for input and how it is processed.	●●●● High. The service provider decides on what data is used for input and how it is processed.
Control on use	●●●○ Low to medium. Once delivered to the purchaser, the data can be used for any application. Terms and conditions may exclude certain types of use.	●●●● High. Since no raw data is provided, the user can only use the service as is intended by the service provider.
Examples	Thousands of companies, like Bloomberg, Equifax, Dun & Bradstreet, Acxiom, etc.	Weather forecasts, navigation service providers, Generative AI tools, etc.



Implementation roadmap

What approach will be used for realizing and implementing the data exchange?



An implementation roadmap outlines the steps the data cooperation should take to accomplish the shared goal and objectives. Like Rome wasn't build in a day, a data cooperation needs to grow step by step.

In this typical implementation these steps are taken along six axes, each of them representing an important aspect of implementation: financial, participation, use cases, technology, organization and legal.

For five phases the diagram shows the progress on these six aspects. From 'Proof of Concept' to 'Scaling up'. This typical implementation roadmap is based on the *5-phase development & implementation method* you'll find in chapter "Change management".

Action:

Define and name the steps that need to be taken to start and scale the data cooperation. Describe the objective and key results for each step.

Additional:

For each step of the implementation plan, list what resources are needed and what activities need to be done. Assign one or more key partners to each resource and activity.

Typical implementation roadmap

In five phases, the data cooperation is build up along six aspects. Starting with an exploratory stage with initial small investments, 2 participants, 1 simple use case feature on small pragmatic technology. Next steps will progress along these axes, the arrows representing an indication of the amount of resources needed to progress.

Phases

- 1 Exploratory stage**
Developing the concept
- 2 Preparatory stage**
Proving the concept
- 3 Implementation stage**
Testing the market
- 4 Operational stage**
Adding value
- 5 Scaling stage**
Growing the number of participants

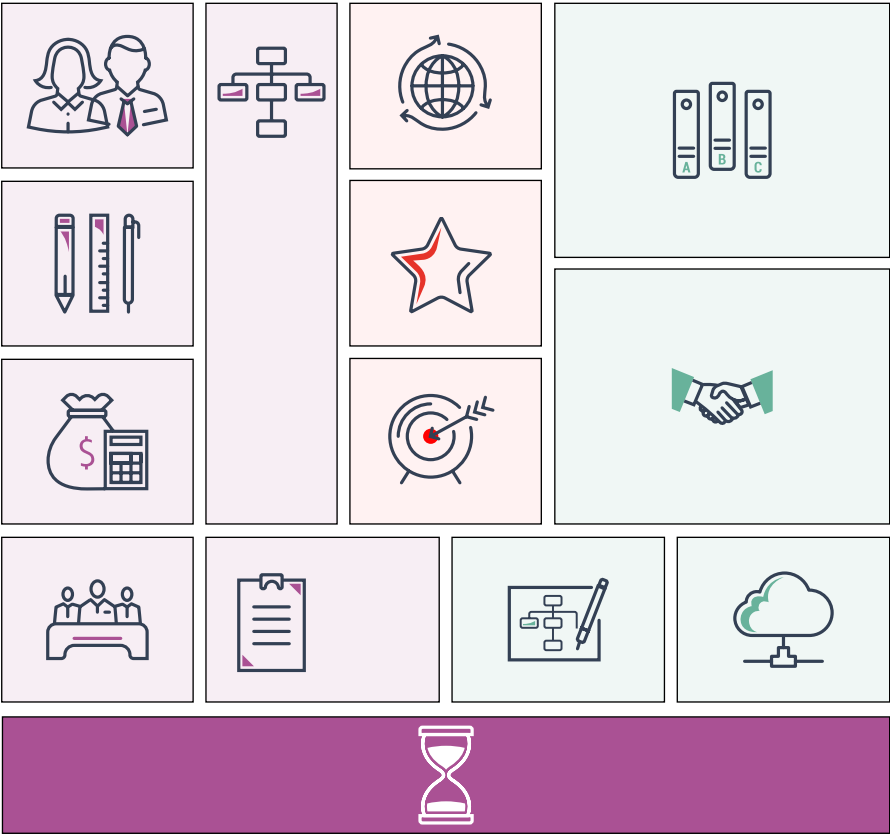
Progress on six aspects per phase



Current status



What is the current status of the cooperation?



This part of the canvas specifies the current status of the cooperation. Especially for external partners it is helpful to understand whether the data cooperation is still exploring its opportunities or it is already a fully functioning data machine.

 **Action:**
Determine the current stage of the data cooperation, using the descriptions on the right.

Stages of the current status

Exploratory stage

One or more (potential) participants are exploring the data cooperation, filling in the data cooperation canvas, attracting other key partners, exploring the legal/financial/organizational possibilities, trying to find funding, etc.

Preparatory stage

Two or more partners have agreed on a viable data cooperation and are working together to prepare all organizations for the implementation stage. Agreements need to be signed, business cases need to be set up and technical implementation partners need to be consulted.

Implementation stage

The data cooperation is being build. Agreements are signed, everyone is on the same page and one or more participants and/or external parties are building the technical parts of the data cooperation.

Operational stage

The data cooperation is operational. Data is flowing and participants are meeting their objectives for the data cooperation.

Scaling stage

The operational stage was successful and now the data cooperation is scaling up in number of participants, number of data sets, features and/or geographical scope.



Data and data sources



What data is exchanged? What data sources are used?



A Data demand & supply matrix can be used to describe what data is required/ demanded by each participant, and what the current availability is for this data set.

Action:

For each key partner, determine their data demand for the data cooperation. This can be must-have essential data sets, could-haves or nice-to-haves (use 'priority' to create this distinction). List these demands in a data demand & supply matrix, like the example shown on the right.

For each data demand, investigate the current availability of the data and/or what key partners or external parties can, could or should provide this data. And what steps need to be taken before the data demand can be met.

On the one-page canvas, the complete matrix may not fit. In this case, list the most important datasets on the canvas and use the matrix as an additional document.

Additional:

Use the combination of priority and current status to create a prioritization for the data cooperation: start with the high priority datasets (first the green ones, than the oranges), continue with medium priority (first green, than orange). Finally, include only the green low priority datasets.

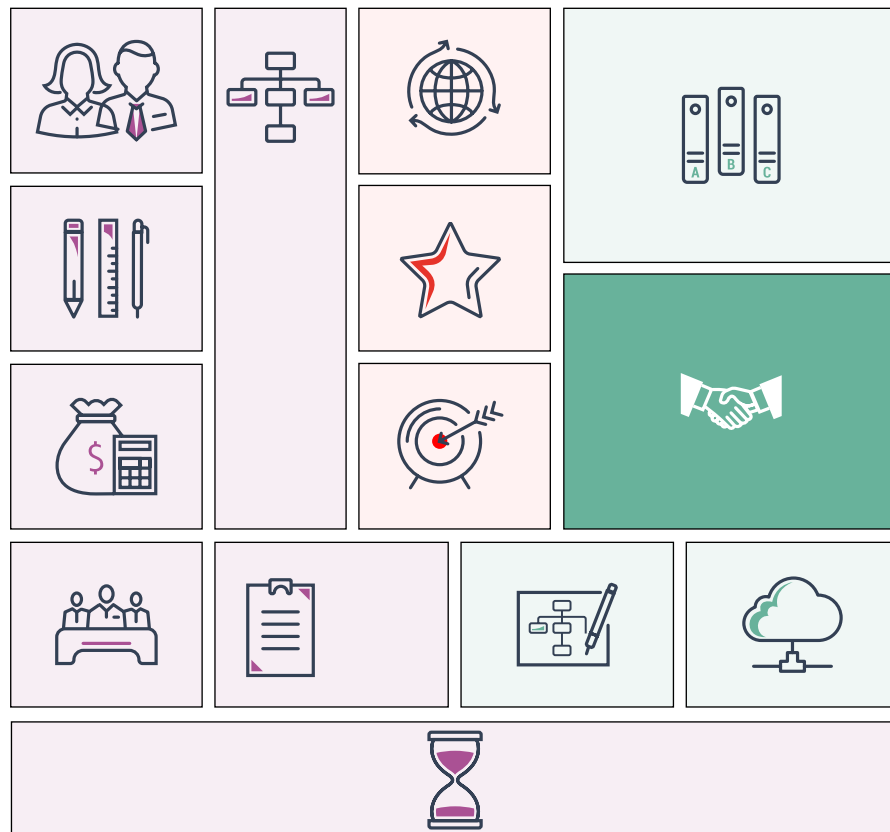
Example Data demand & supply matrix

Key partner	Data demand	Priority	Data supply / availability	Current status
Participant A	Data on road network	☆☆☆	Data available as open data by Participant B	● Available. Use api.road.network/2.0
Participant A	Data on actual driving speeds on road	☆☆	Data available from Participant B as soon as contract is signed	● Available if contract is signed
Participant A	Real-time floating car data	☆	Participant C can provide FCD with 1-day delay. Real time data requires expensive contract.	○ Start using historic data. Assess expensive contract later
Participant B	Feedback from service providers	☆☆☆	We will need to convince service providers to join or data cooperation	○ Requires effort to convince service providers
Participant B	Personal data of road users	☆☆	Data can no be shared because of GDPR. May be it can be aggregated, but this requires major effort from Participant D	● Will never be available.
Etc.				
● Not (yet) available. Will never be available or requires major effort ○ Not yet available, requires medium effort ● Available or available soon with minor effort				

Interoperability



How can the data be uniformed/standardized/combined? What shared concepts, languages, formats, or methods can be used? Is it hard to combine all the data? Or are standard definitions available? What data standards & formats are used or need to be used?



The current level of interoperability defines what additional effort is needed to compare, combine and aggregate data from different providers. If standard definitions are used this is probably fairly easy, while it might be a difficult job if there are no standards or shared understanding.

The availability of standards and the extent to which they are applied are a key success factor for data cooperations.

Action:

On a high level, use a couple of bullet points to cover the essentials of interoperability of the data cooperation. For example, mention the existence or absence of (inter)national standards, whether all partners use the same standards or formats, whether they use existing (inter)national aggregators or data providers, what effort is needed to convert/map data or definitions, etc.

Additional:

For each dataset in the data demand & supply matrix, specify the current level of interoperability, the available standards and methods and what effort is required to reach a satisfying level.

Typical elements of interoperability

Current level of interoperability

1. No standards or shared understanding
2. Shared understanding
3. Ad-hoc standards/definitions that can be mapped
4. Local standards that are defined and need to be mapped
5. Standard definitions available (worldwide)

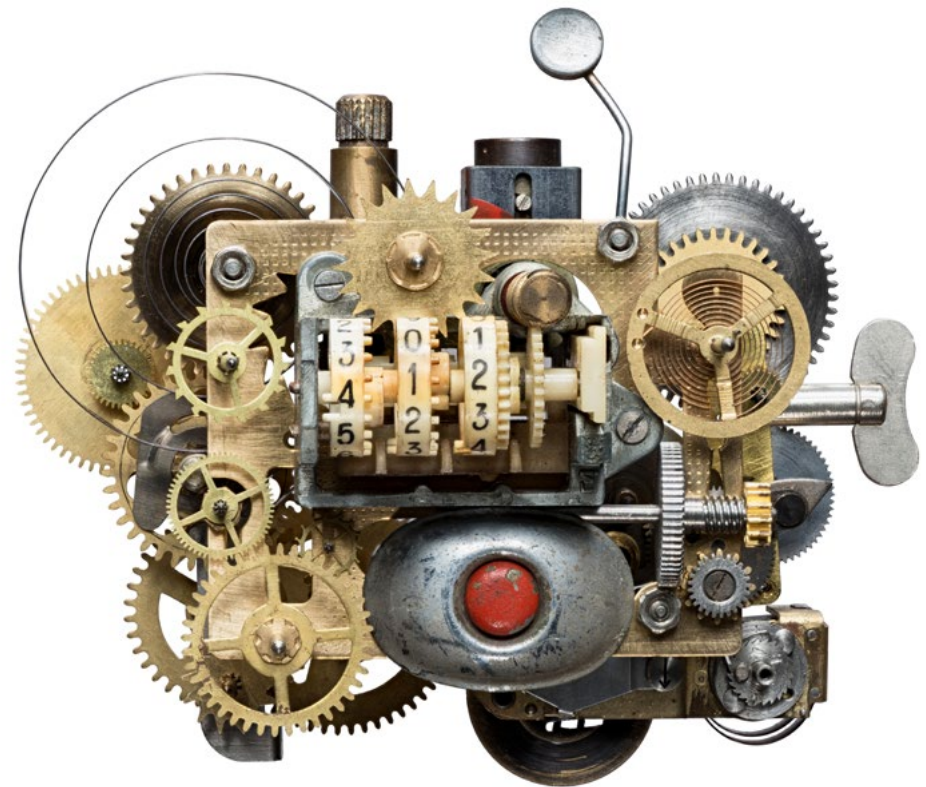
.....

Available standards, concepts, languages, methods

.....

Effort required to reach a satisfying level

.....



Technical concepts/models



What technical concepts or models need to be in place for the data exchange. What MIMs are implemented and how are they implemented?



In this canvas area, describe what technical concepts and models are or will be used for the data cooperations.

For cities and communities, OASC (Open & Agile Smart Cities) has developed 10 Minimal Interoperability Mechanisms (MIMs), a set of practical capabilities based on open technical specifications that allow data cooperation participants to replicate and scale solutions globally.

Use these as reference, or name the technical concept/models yourself.

Action:

List all MIMs and other technical concepts or models that are or will be implemented in the data cooperation.

Additional:

For all items of this list, describe how they will be implemented and how this effects the data cooperation and the required effort/resources.

MIMs & other typical concepts/models

1. What MIMs are/will be implemented in the data cooperation?

How will the be implemented?

Read more about MIMs at <https://mims.oascities.org/>

1. MIM1 – Context
2. MIM2 – Data Models
3. MIM3 – Contracts
4. MIM4 – Trust
5. MIM5 – Transparency
6. MIM6 – Security
7. MIM7 – Places
8. MIM8 – Indicators
9. MIM9 – Analytics
10. MIM10 – Resources

2. What other technical concepts/models are/will be implemented?

For example (a non-exhaustive list of examples):

- APIs (Application Programming Interfaces)
- Data Mining
- Data Warehousing, Data Lakes and Lakehouses
- Data Integration and ETL (Extract, Transform, Load)
- Predictive Analytics
- Machine Learning Models
- Natural Language Processing (NLP)
- Deep Learning
- Data Governance and Privacy Frameworks
- Data Visualization Tools
- IoT (Internet of Things)
- Blockchain Technology
- Federated Learning
- Parallel Computing
- Cybersecurity Measures
- Semantic Web Technologies
- Data Versioning and Lineage Tracking
- Etc.

Infrastructure characteristics



What technical infrastructure is needed for the data exchange?



A complete technical infrastructure architecture or blue print may be to complex for the data cooperation canvas.

Instead we ask to describe the technical infrastructure in only a few sentences, guided by the typical questions provided here.

Action:
Describe the current or needed technical infrastructure of the data cooperation, using the typical questions and example on the right.

Additional:
If available, a high level architecture blue print of the infrastructure can be added to the additional documents.

Typical questions & example

Typical questions / technical infrastructure characteristics

- What cloud/server infrastructure is used?
- What technology stack is used?
- What standard software is used?
- What (cyber)security measures are taken?
- In-house development or external parties?
- Central/decentral/distributed model
- How can be connected (API, feeds, downloads, etc.)?

Example technical infrastructure characteristics

- Our data cooperation is using Amazon Web Services (AWS) for scalable storage and computing capabilities. The technology stack includes PostgreSQL, Apache Hadoop and code in Python. We're using standard software like Tableau or Power BI for visualization and reporting.
- Development is done by a mix of in-house development and collaboration with external parties, utilizing specialized expertise where needed. The model is typically decentralized, allowing various stakeholders to contribute and access the data while maintaining centralized governance and security protocols.
- Connection to the infrastructure is established through APIs for real-time data access, feeds for continuous data ingestion, and regular downloads for periodic data updates, ensuring seamless and efficient data exchange within the cooperation.

TOWARDS A PROFESSIONAL DATA COOPERATION

OK, the Data Cooperation Canvas is filled in. And now what?

In this chapter we describe a path for change management towards a professional data cooperation. Since it is impossible to create a generic development & implementation plan for all data cooperations, we use a typical approach for setting up a professional data cooperation on a European level with potentially hundreds of participants:

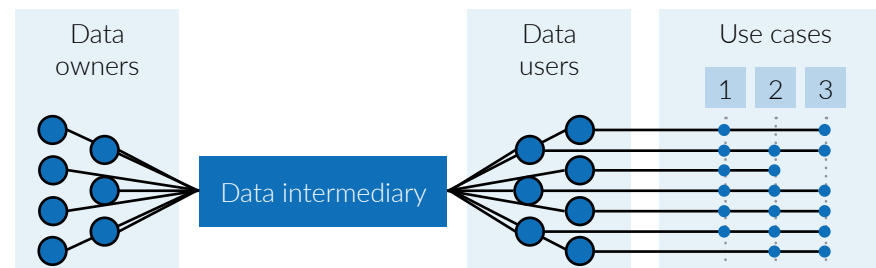
The Data Intermediary

In the 5-phase development & implementation method we will describe a special organization within a professional data cooperation, the 'Data intermediary'.

This Data intermediary will receive the data of multiple (tens, hundreds or even thousands) data owners, process it in some way and provide services to multiple (tens, hundreds or even thousands) data users.

The Data intermediary will start with 1 service or use case, which will be extended in the 5 phases.

A Data intermediary may operate at a local or regional level. But for this example we assume that the Data intermediary functions at a national, or, preferably, European level.



Envisioning our goal

What does this Data Intermediary look like?

To determine the strategy, it is important to start by envisioning the final goal: A professional data cooperation, providing actual solutions for many, with a professional service proposition, delivered by a professional data intermediary organization.

Providing actual solutions for many

- A cooperation of many (tens, hundred or thousands) participants sharing data on national or European scale
- Where new participants can easily join under clear access conditions, regarding technical, legal and financial aspects

With a professional service proposition

- Providing great data availability, enabled by the participation of essential players in the sector
- Professional technology based on industry standards that supports seamless data exchange
- Clear rules to participate in the data exchange
- Clear cost structure for on-boarding and participation

Delivered by a professional data intermediary organization

- A mid-sized, not-for-profit (see the Data Governance Act) organization with:
 - Senior management
 - Professionals for market development, user relations and sales.
 - A strategy for future product development including user representation.
 - A professional IT team, not locked-in by external providers
 - A full-scale help desk
- A balanced governance structure where all user groups are represented.
- A serious legal basis with a service level agreement (SLA) and data sharing agreements.
- Financially stable and not depending on external funding.

The 5-phase development & implementation method

To realize a professional data cooperation like we envisioned, we propose a 5-phase development & implementation method. Starting from scratch, we will work towards our described goal in 5 phases.

Principles

To realize a successful data cooperation, we advise the following development & implementation principles:

- Think big, start small; think global, act local.
- Work agile, bottom up, from small to big, upscaling is standard.
- A Data Intermediary is not-for-profit initiative (rules of the European Data Governance Act!)
- Work on interoperability; synergy with other data cooperations (also from other sectors) will add a lot of value.
- Big participants should take the lead, make it easy for small participants to join; many small participants will add a lot of value for big participants.

In the next pages we will describe the 5 phases that take these principles into account to implement the envisioned professional data cooperation:

- **Phases 1 and 2**

The startup data cooperation based on an MVP

- **Phases 3 and 4**

Start of the data intermediary as a professional organization

- **Phase 5**

Scaling up a completely implemented data cooperation

Preparations (could be considered Phase 0)

Goal of this phase is to enable the initiative to start as soon as possible with minimal means.

- **Concept**

A first vision is developed:

- A use case is carefully selected in which a real problem is solved, without any legal issues.
- A flow chart is designed between participants and data intermediary that explains the processes between parties.

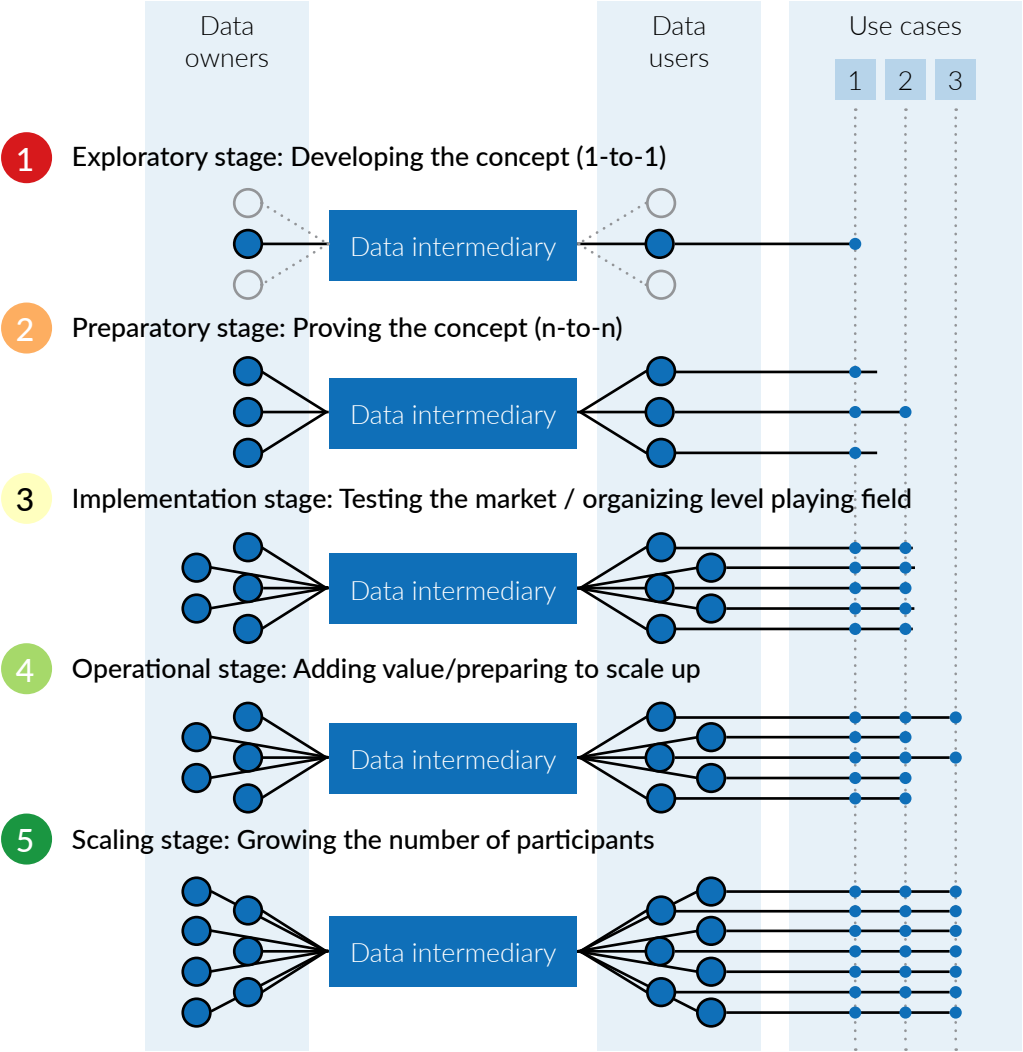
- **Participation**

Sector organizations and/or a small group of potential participants form a pilot group to take the initiative to start a new regional/national/ European data cooperation. These “initiators” start a not-for-profit cooperation that will develop into a future Data Intermediary.

- **Financial**

A limited (mostly local) budget needs to be found by the pilot group.

The 5 phases in one overview

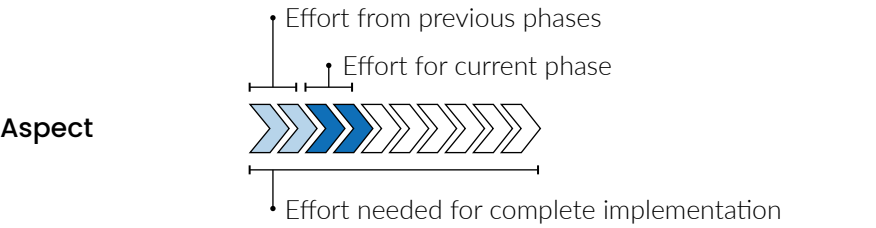


Situation & activities

On the next pages, we will explain the situation and activities for each phase.

Focus & progress

For each phase, we have visualized the focus and progress on six aspects. It shows the results of the previous steps and the current focus on a scale from zero to complete implementation:



In this way, we will show the focus and progress for these six aspects:

- Financial
- Participation
- Use cases
- Technology
- Organization
- Legal

Please see page 56 for more details on these aspects.

1 Phase 1: Exploratory stage

Developing the concept

Goal of this phase

Test the concept in a one-to-one exchange, while potential other participants are involved from the start. By realizing a Minimum Viable Product (MVP) in a simple, pragmatic organization.

Situation & activities

The pilot group details the concept of the cooperation by filling in all aspects of the Data Cooperation Canvas.

Financial

The (local) budget for this phase is very small and probably funded by the initiators network. At the end of the phase the budget for phase 2 is prepared.

Participation

A small group of “representative participants” (big and small, regional spread, etc.) is carefully selected and form a pilot group. With “small” we mean 6 – 10, as less as possible. The goal of phase 1 and 2 is to prove that a valuable use case really works and adds real value. Not to prove all use cases in all situations. Work bottom up. Important is that this group shares the vision behind the concept of the data cooperation as described in the Data Cooperation Canvas.

Use case

Make a definitive choice on the first use case. Avoid any legal issues and check the data availability and data quality of the participants in the pilot group that you will need for the use case.

In the selection of the use case, it is crucial that the data generated by the use case can be directly applied in an existing or new application, providing immediate added value. Ideally, this should also result in added value for the participant providing the data.

Organization

If possible, the pilot group forms a startup (as a legal entity) for the data intermediary. If not, they look for a practical solution that can play this role temporary. In case privacy sensitive data is involved, parties should be extra careful in this situation.



Technology

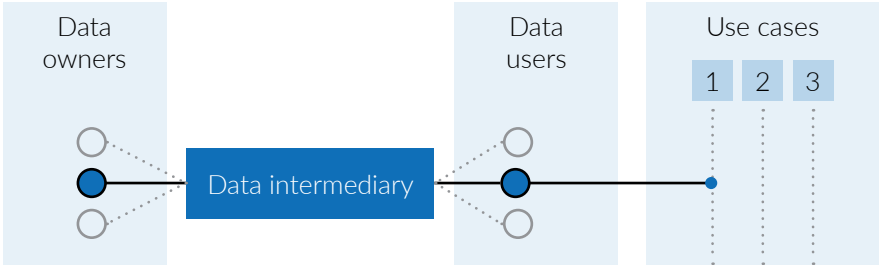
Based on the use case, the concept and the canvas the startup Data Intermediary develops a Minimal Viable Product. An initial one-to-one connection is created between two parties from the pilot group, using a Data Intermediary. The other participants are involved from the beginning. They will review the solution, and commit to connect to the MVP in phase 2.

Legal

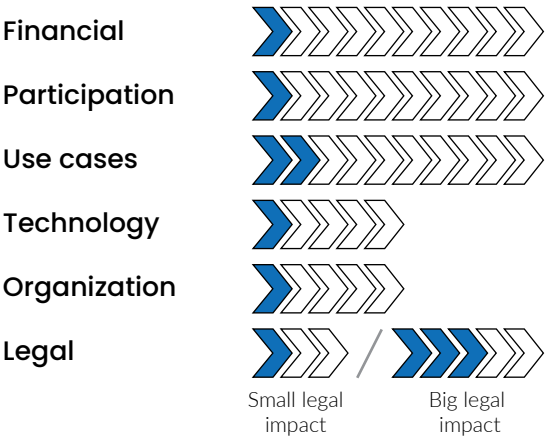
In case severe legal limitations (like GDPR) are relevant for the selected use case(s), this will have a big impact. It completely dominates the concept of the cooperation. In that case the legal focus in phase 1 will be intensive. In more open data cooperations, the legal focus will be average.

In general: agreements between the members of the pilot group and the (temporary) data intermediary are legally signed. For this non disclose agreements and data sharing agreements are developed and signed.

In case data owners are not willing to share relevant data, the Data Act might help. But only as a last option. Experienced learned that in these cases an innovative approach is much more effective than a legal approach.



Focus & progress



2 Phase 2: Preparatory stage

Proving the concept

Goal of this phase

Proving the concept of the data cooperation in multiple situations and finding new investors. Preparing for the "big leap" of phase 3.

Situation & activities

Financial

Activities are financed by a small extra (local) budget, probably funded by new active data owners and data users in the pilot group. Most important in this phase is to find investors for the big budget for phase 3. These parties probably operate on regional, national or European level (depending on the scope of the data cooperation).

Use cases

From contact with potential investors of phase 3, extra use cases might pop up. It is possible that the funder of phase 3 may impose conditions on the use cases. Take this into account during phase 2.

Participation

No new data owners and data users are introduced.

Technology

All participants from the pilot group that were not connected in the previous phase are now joining. They connect to the MVP, which may require minor adjustments as a result.

For the new use cases, the MVP is extended. Preferably all data owners and data users of the pilot group will implement these extra use cases.



Organization

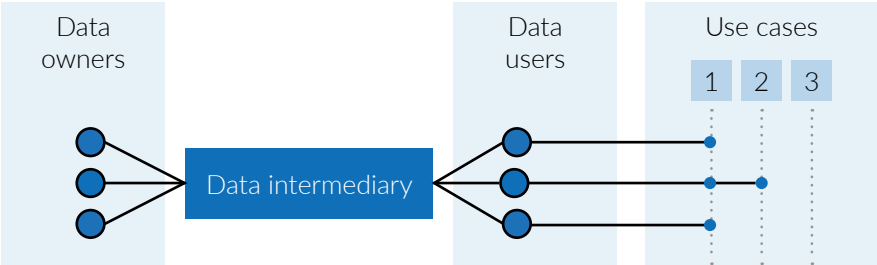
Not much is changed in this field compared to phase 1.

Legal

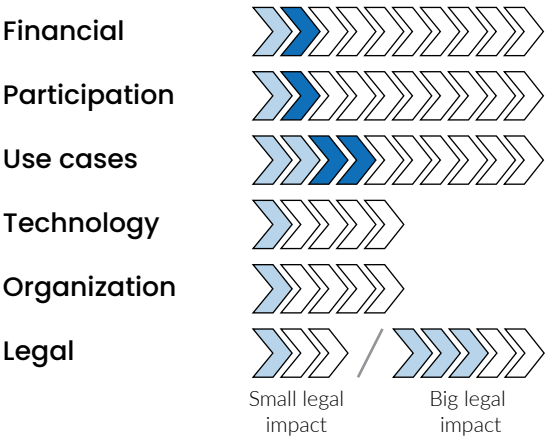
Non-disclosure agreements and data sharing agreements are signed by all data users, data owners and the temporary data intermediary.

At the end of phase 2 the effectiveness of the concept is formally reviewed with candidate investors on regional, national or European level of phase 3 (and 4). This includes goals for future phases.

Phase 2 is finished by signing a contract with these investors.



Focus & progress



3 Phase 3: Implementation stage

Testing the market / organizing level playing field

Goal of this phase

Make a formal start with the data intermediary based on state-of-the-art technology and find out if there is a fit to the market. In case of a B2G cooperation, a level playing field is organized.

Situation & activities

Financial

For this phase there is a considerable budget, probably from regional, national or European investor(s). At the end of phase 3, you compile a complete business case including a detailed overview of the investments required for phase 4.

Participation

This is the right moment to approach the most sought-after participants to gauge their interest and adjust your approach accordingly. This will be

included in the marketing plan (see 'Organization').

New participants can also be added for political reasons (for instance at the request of the investors).

In Business to Government (B2G) cooperations, this phase is used to create a level playing field. All (relevant) business participants are invited to join.

Use cases

No new use cases are introduced in this phase.

Organization

In the beginning of phase 3 a governance board of the Data Intermediary is formed, which represents all types of users.

This phase is also the start of building the Data Intermediary as a formal and professional organization. Small teams for technology, market- and product development are set up and a basic help desk is installed, all supported with professional tools.

The market development team will set up a strategic marketing plan which contains:

- A plan for growing the data cooperation including high potential data owners and data users, a wish-list for extra new use cases, etc.
- A detailed design of the service offering of the data intermediary.
- A first policy for on-boarding and participation of the data cooperation during the coming phases.
- A plan for promoting the activities of the Data Intermediary.

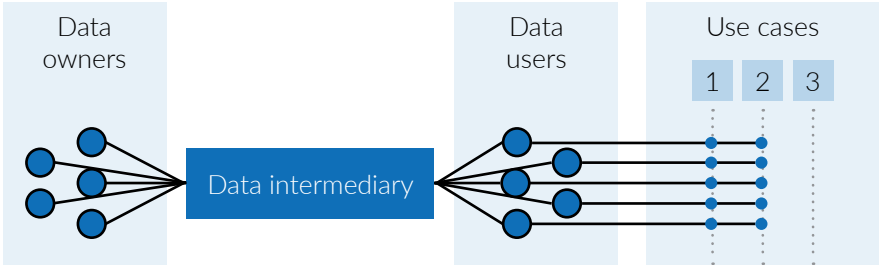


Technology

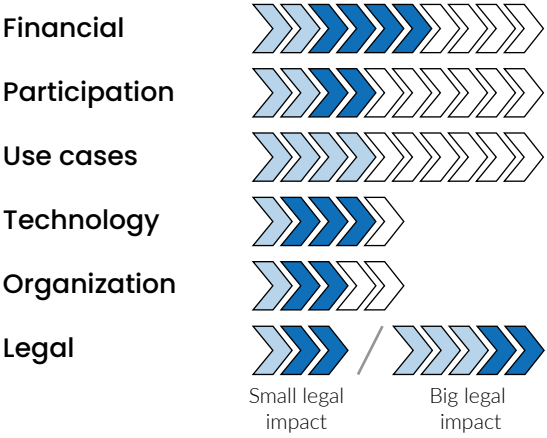
The technology of the MVP needs to be completely rebuild to meet the requirements for phases 3 to 5. Transferring the MVP to the final technical infrastructure based on state-of-the-art-technology is the main activity for the IT team of the formal data intermediary. New product and service providers might be selected and contracted. Important is that dependency on third parties is as low as possible, by using as much open standards, open source software and open contracts as possible. While designing and implementing the technology, there should be a strong focus on meeting the required scale and performance of the next phases.

Legal

Based on the designed service offering, professional Terms and Conditions needs to be developed. In this phase, we will develop the final Data Sharing Agreements and Service Level Agreements that will be utilized up to and including phase 5. All documents will be signed with all participants of phase 3.



Focus & progress



4 Phase 4: Operational stage

Adding value / preparing to scale up

Goal of this phase

Prepare the data intermediary for the open market introduction. This requires a professional not-for-profit organization with an interesting service offering, operating in an interesting and fast-growing network of data owners and data users.

Situation & activities

Finance

This is the last phase to use external funding. From phase 5, the data intermediary will be paid for by all participants. Since investment must be made in technology and organization, the budget for this phase needs to be considerable. The results from phase 1, 2 and 3 must be impressive enough for the investors to take this last risk.

Use cases

A new set of extra use cases is carefully selected. The goal for this is to make paid participation for new participants at the start of phase 5 as attractive as possible. Information on which use cases to develop, comes from the experiences during testing the market in phase 3.

Participation

There is no focus on adding new participants in this phase.

Organization

In phase 5, a significant number of participants must be efficiently on-boarded and serviced. Therefore, the organization and processes must be prepared and professionalized during phase 4. Supporting systems should be selected / developed in a way that scaling of the processes is relatively easy. After finishing and testing,



the new use cases are introduced and promoted to the current participants. This may need launching a marketing/communication campaign towards potential participants.

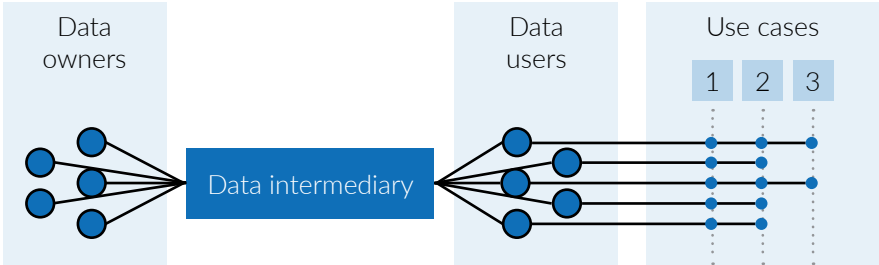
Technology

As the new use cases are beyond the scope of the MVP, new technology and software development are needed in this phase.
Working on scalability of the technology will be crucial in phase 5. This applies to the primary process as well as to supporting systems such as billing, administration, etc.

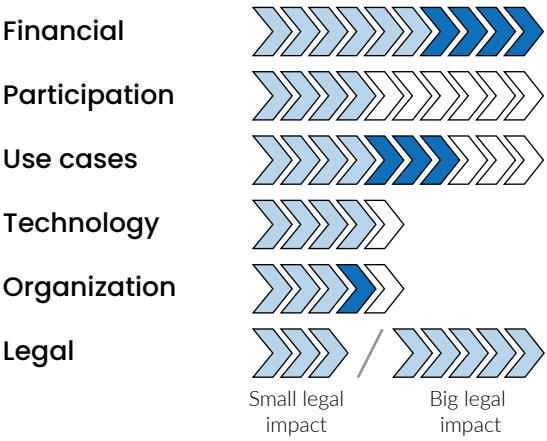
Legal

The details of the Terms and conditions needs to be reviewed and adjusted.

At the end of phase 4 the data intermediary is financially independent of external funds, with a balanced budget. It is now grown to a small-sized professional organization with attractive services package, including a professional SLA and clear conditions for new participants to join the network on all legal, technical, procedural aspects.



Focus & progress



5 Phase 5: Scaling stage

Growing the number of participants

Goal of this phase

Making the transition from external funding to financial self-sustainability. This requires professional planning within the limits of available budget. The data intermediary will smoothly grow as an organization and its services.

Situation & activities

Financial

“Not-for profit” is the rule. All income of the data intermediary will be invested in better services for its users. At the start of phase 5, the income may be modest; to sustain the organization, it is essential to maintain a certain level of working capital.

The management team and governance board is in the lead here. When the number of participants grow, the rates might (slowly) drop.



Participation

From the start of phase 5 the data cooperation is open for participation. To be effective it will need a boost right from the start. This should be very accessible:

- Clear data / technical standards
- Clear legal conditions
- Standard procedures for on-boarding
- Clear costs for on-boarding and participation (per year)

Organization

Only quick growing numbers of participants can cover the costs of the professional organization. A combination of smart marketing, an efficient process, and low connection costs is essential.

The growth of the data cooperation will come from the attractiveness of its services.

This attractiveness is depending on:

- The field of expertise
- The first participants and the data that is available
- The first use cases
- The data that is shared
- The costs for on-boarding and participation (per year)

As participation grows, all processes of the Data Intermediary will grow with it. This is possible because budgets are increasing:

- The help desk will grow its capacity and opening hours
- Out of user contacts and user representation ideas for new use cases are born. A formal process is in place to review, select and implement new functionality.

Use cases

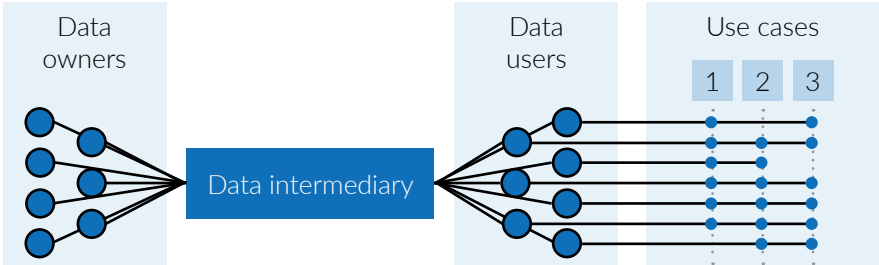
In the beginning of phase 5, no new use cases are developed or implemented. There is only room to diversify, as soon as there is sufficient budget available for it.

Technology

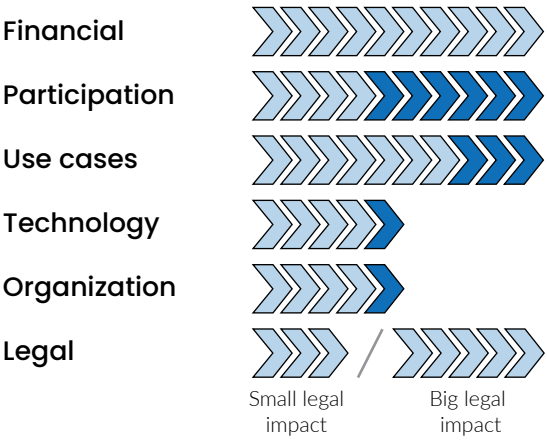
Soon after the participation accelerates, scaling of technology is critical. When it comes to the budget, the first priority is to scale up, to ensure the SLAs can be guaranteed. As soon as participation and budgets grow there will be room for further investments to implement new functionality.

Legal

No changes from phase 4. Only the number of agreements will grow.



Focus & progress



The 6 aspects of implementation method

For each step we have visualized the activities and progress on six aspects. On these pages we provide our lessons learned for each of these aspects.

The first three aspects are driving: they steer towards the objectives of the data cooperation. The next two, technology and organization, are adding value by supporting these objectives but also affect them from their influence on the budget. The latter, legal, is just following the requirements of the other aspects. Unless serious legal limitations (like GDPR) are restricting the use cases.

Driving aspects

Financial

Experience learned that budgets have the biggest impact on composing phases 1 till 5. In most cases the start of a Data Intermediary is depending on local or regional budgets. Financiers must be convinced of the fact that the scale of national / European data cooperations will add a lot of value to local / regional initiatives. Funds from the Digital Europe Program can help in these situations. Critical: during the design of the goals of each phase the project manager has to envision which party/parties will be the financiers of the next phase and what deliverables will be needed to convince them to invest.

Participation / market

Participants are most important in a data cooperation. A diverse and extensive participant base not only enhances its appeal to newcomers but is also crucial for the overall community. Participation of certain partners may play a critical role in the entire community's success. Therefore, meticulous selection of the initial partners during the establishment of a new data cooperation is essential. Strive for a balanced mix of both prominent and smaller-scale data owners and users, taking into account regional or national variations. Opt for data owners contributing compelling, high-quality data and data users capable of deriving substantial added value through the integration of this new data into existing applications. Choose partners strategically to attract new and noteworthy participants.

Use cases / Data selection

The choices around the first use cases are also critical:

- What data will owners be willing to share?
- What is the added value for the use cases for data users?
- What are data users willing to do / pay for this?
- Will there be enough data users?

These questions are crucial for the selection of use cases, in phases 1 and 2 to validate the concept and convince investors for phase 3, and in phases 4 and 5 to be successful in the open market.

Balancing aspects

Technology

Due to limited budgets in phases 1 and 2, a Minimal Viable Product (MVP) is crafted. Reliance on third parties is not a concern during these stages. The MVP may be discarded after phase 2, to start again from scratch in phase 3. From then on, the definitive technical infrastructure is established, emphasizing the importance of a clear vision for the technical future and ensuring independence in technical, organizational, and legal aspects.

The MVP phases allows for valuable experience to inform optimal decisions at the onset of phase 3. Regardless of circumstances, adherence to open industry standards is highly recommended, starting from phase 1.

Organization

Working for a new Data Intermediary very much feels like a regular tech start-up. The operational style is entrepreneurial, innovative, experimental, taking balanced risks. The motivation is not driven by personal wealth aspirations but by the pursuit of a leadership position in 'a cooperative world' since the endeavor is structured as a not-for-profit. Idealism is integral to the journey. In phase 1 and 2 the team is small; budgets are (very) limited. But founders could grow towards a role in the future management team of the full-grown data intermediary in phase 5. From the start of phase 3 the Data Intermediary will grow rather fast. At the start of phase 5 the organization needs to be mature.

Balancing / following aspect

Legal

Legal limitations (like GDPR) will be relevant for quite some data cooperations. In those cases, this will have a big impact on the concept, which will be determined in the set up of the cooperation.

After that, agreements made during a phase between the participants themselves and the data intermediary are legally formalized. Data sharing agreements are developed and signed, often accompanied by Non-Disclosure Agreements.

Legal is always following the innovation process. When legal activities would take the lead, things would get formal too quickly; it would paralyze the innovation process.

AN EXAMPLE CANVAS: IDEA (The Netherlands)

Now we have covered all the elements of the Data Cooperation Canvas in detail, we will provide a real world example of a Data Cooperation Canvas, regarding a Dutch data cooperation between local authorities, a national data warehouse and commercial navigation service providers:



Validating planned road works using real-time data

On a national level, local authorities cooperate to provide open data on road works. This open data is based on the planning processes of local road authorities, like the City of Amsterdam. This plan data may differ from the actual data, since the actual work has been outsourced to various subcontractors. Despite many efforts, these subcontractors don't update the city on their real-time status.

Enter IDEA (Intelligent Data Exchange Alliance), a data cooperation that validates data of planned roadworks and closures based on real-time floating car data (FCD) from navigation service providers. Based on the real-time, actual position and speeds of vehicles, IDEA detects most road works and closures within 10 minutes.

For navigation service providers, this results in high-quality, validated, real-time data. For road authorities, anomalies from the planned data are visible.

On the right, data cooperation is described using the Data Cooperation Canvas, as an example of how to use the canvas.

Organizational

Key partners

- NDW (National Datawarehouse on road traffic)
- City of Amsterdam, Traffic Department
- City of The Hague, Traffic Department
- Province of North Holland
- Rijkswaterstaat (National Road Authority)

Resources

- A national platform/database on mobility data (including floating car data) is available (NDW)
- A coalition of data science developers associated with the NDW

Business case

The road authorities invest in IDEA to create high quality data. This data will improve the information to road users (through the service providers) and may be used to efficiently control subcontractors. It is financed by all road users in an existing nationwide cooperation (NDW).

Governance model

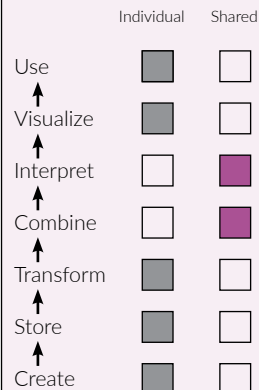
IDEA is open to all road authorities in The Netherlands. The resulting data feed is available for free for all service providers.

NDW is technical lead, and through its member structure, a steering committee, represented by all key partners, makes decisions about IDEA.

A user group is being set up to govern the functional parts

Shared processes

IDEA combines all data and adds a validation status based on the real-time actual traffic situation. Each partner can visualize and use this information. Navigation service providers, for example, will bring this to road users.



Why?

Context

Road authorities (local and national) have open data on road works. This data about the planned road works may differ from the actual road works due to f.e. subcontractors. The actual period of delay of closure due to road works could be hours or days off from the open data.

Added value

Service providers and road authorities want to have data on **actual** road works.

By validating the planned road works, using live data (from floating car data (FCD)), IDEA generates a high quality, real-time data feed for road works.

Motivation & objectives

Providing high quality, real time data on road works. **Service providers** can provide better information to road users. **Road authorities** have insight into their road works' actual impact. For example to check on subcontractors.

Technical

Data & data sources

Input data

- Planned dates and details for road closures and construction works from local, regional and national road authorities.
- Floating Car Data from one national service provider
- Feedback data on validated IDEA-data from service providers

Output data

- Validated high quality data for road closures and construction works

Interoperability

- All road authorities enter their planned data using only a handful of software, which is aggregated on a national level by NDW and converted to the DATEX-II format.
- All road authorities and service providers are familiar to the DATEX-II format, an European standard for road- and traffic related data.
- Because of using different map, roads and road segments may need to be mapped by the users of the IDEA-data, based on the geometry.

Implementation roadmap

1. Local pilot, directly based on the national platform, so a nationwide implementation is (technically) easy
2. Create sponsorship on a national level (organizational + financial)
3. Deploy IDEA for all road authorities in the Netherlands

IDEA is currently preparing for phase 3

Technical concepts/models

- The validation algorithm is using machine learning
- Input and output feeds are using simple REST APIs
- Identity Management is done by NDW using Keycloak

Infrastructure characteristics

- A central cloud solution is used (Microsoft Azure infrastructure)
- Python is used as the main language for development, while in the scaling stage parts will be ported to Java
- Development is a cooperation between in-house development teams and external companies

Current status

IDEA is in the *Operational Stage*

Exploratory stage

Preparatory stage

Implementation stage

Operational stage

Scaling stage



Data Cooperation Canvas Template

At the right, an empty Data Cooperation Canvas is available. Print or copy this page and get started with the Canvas!

Download a digital version at <https://www.datacooperationcanvas.eu>



Organizational

Why?

Technical

Key partners 	Shared processes Individual Shared Use ↑ Visualize ↑ Interpret ↑ Combine ↑ Transform ↑ Store ↑ Create ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ 	Context 	Data & data sources 	
			Interoperability 	
Resources 	Added value 	Motivation & objectives 	Infrastructure characteristics 	
Business case 				
Governance model 	Implementation roadmap 	Technical concepts/models 		
Current status What is the current status of the cooperation? ☐ Exploratory stage ☐ Preparatory stage ☐ Implementation stage ☐ Operational stage ☐ Scaling stage 				

ABOUT THE AUTHORS

Meet Ron and Jasper, seasoned experts with a collective experience of over 20 years in business consulting and project management. Throughout our extensive careers, we've actively engaged with numerous (potential) data cooperations, each presenting its unique challenges and opportunities. Despite their diversity, we've identified consistent key success factors.

The creation of the Data Cooperation Canvas marks a pivotal moment for us—a tool that amalgamates our experience and insights. We're enthusiastic about sharing this comprehensive framework with you, a result of our eagerness to contribute to the community.



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Specialist Smart City development, open and shared data, public private partnerships, city strategic partnerships (f.e. Google and TomTom), member team Chief Technical Officer City of Amsterdam.



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Braxwell, innovation partners

We'll help improve your organization. Based on a well thought-out strategy and a realistic roadmap. We like to think things out, but prefer action over reports, our feet grounded by practical experience.

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strategy



plan



go

